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A mechanism-guided, low-input Lactiplantibacillus plantarum strategy for attenuating antibiotic resistomes under oxygen re-entry during anaerobic fermentation

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ABSTRACT

Oxygen re-entry during the handling of anaerobically fermented biomass is accompanied by changes in microbial communities and antibiotic resistance profiles. This study evaluated the effects of different lactic acid bacterial inoculants on the attenuation of antibiotic resistance genes (ARGs) in cattail silage during aerobic exposure after anaerobic fermentation. Using shotgun metagenomic sequencing, we tracked ARGs, mobile genetic elements (MGEs), and microbial succession in the control (CK), *Levilactobacillus brevis* (LB), a self-isolated *Lactiplantibacillus plantarum* (LP), and mixed-culture (M) treatments after 60 d of fermentation and 0–9 d of aerobic exposure. The resistome was dominated by glycopeptide- and tetracycline-resistance genes, with tetM and vanY as the major subtypes. Throughout the exposure period, LP maintained the lowest ARGs abundance, whereas LB enriched tetracycline resistance and the mixed culture did not provide additional ARGs attenuation. In all treatments, MGEs, mainly integrase- and plasmid-associated markers, covaried with ARGs, and their mobility potential gradually decreased during prolonged exposure. Aerobic exposure also induced clear microbial succession, including the disappearance of major ARGs hosts and the late enrichment of yeasts such as *Pichia* and *Candida*, which were negatively associated with ARGs abundance. Network analysis, random forest modeling, and partial least squares path modeling suggested that physicochemical changes induced by oxygen re-entry were linked to microbial community restructuring, accompanied by shifts in MGEs profiles and ARGs dynamics Overall, inoculation with the self-isolated LP strain, together with limiting aerobic exposure time, provides a practical and low-input strategy for reducing ARGs burden and potential mobility risk during the handling of fermented biomass.

1. Introduction

Antimicrobial resistance (AMR) has emerged as a major global threat to human and animal health, and its strong environmental dimension is now widely recognized within the One Health framework [1]. Antibiotic resistance genes (ARGs) and associated mobile genetic elements (MGEs) are increasingly viewed as environmental contaminants because they mediate horizontal gene transfer among pathogens, opportunistic pathogens and commensal microorganisms across clinical, agricultural and natural ecosystems [2,3]. Global syntheses show that ARGs are widespread in soils, waters and other environmental media, with agricultural systems often exhibiting elevated ARGs loads relative to non-agricultural environments [4–6]. Manure production, storage and land application can further mobilise ARGs and MGEs, creating a continuum of AMR risk from farm to the wider environment [7,8].

Within ruminant production, forage resources are commonly preserved through solid-state anaerobic fermentation, particularly in regions with harsh climates or short growing seasons [9,10]. During this anaerobic fermentation process, large amounts of plant biomass are compacted and stored under oxygen-limited conditions, where dense and dynamic microbial communities assemble and drive rapid physicochemical changes (e.g., pH decline and organic acid accumulation). Importantly, oxygen re-introduction is often unavoidable during subsequent handling and utilization, imposing a sharp redox perturbation

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that can trigger community reassembly, organic acid consumption and substrate depletion. Recent work has reported diverse ARGs assemblages in fermented forage matrices and suggested that substrate type, anaerobic fermentation conditions, fermentation quality and management practices can markedly influence ARGs occurrence and persistence [11,12]. Residual materials, leachates, and ARGs-carrying microbes entering manure handling chains may further facilitate dissemination to soils, waters and even plant surfaces, linking fermented forage systems to the feed–animal–environment continuum [13,14]. However, the successional dynamics of resistomes across the anaerobic fermentation–to–aerobic re-exposure transition, and their implications for environmental AMR dissemination, remain insufficiently resolved under practical oxygen re-introduction scenarios.

Lactic acid bacteria (LAB) inoculants are widely used to improve anaerobic fermentation performance by accelerating lactic acid production, lowering pH and inhibiting spoilage yeasts and moulds [15,16]. *Lactiplantibacillus plantarum* (*L. plantarum*) is among the most applied species because of its robust growth and acidification capacity across diverse forages. Beyond agronomic benefits, LAB inoculation has recently been shown to influence AMR-related risks in fermented forage and related matrices by altering the abundance of ARGs, MGEs and potential bacterial hosts [17,18]. At the same time, some LAB strains carry intrinsic or acquired resistance determinants and prophages, raising concerns about their possible contribution to ARGs dissemination in food and feed chains [19,20]. Moreover, indirect pathways mediated by acidification strength, organic acid profiles and oxygen availability may generate stage-dependent impacts on ARGs hosts and MGEs-linked transfer potential, which remain mechanistically unclear.

Cattail (*Typha* spp.) is an under-utilized wetland plant that has been explored as a low-cost forage resource in high-altitude or resource-limited pastoral regions [21]. Its high moisture content, fibrous structure and epiphytic microbiota can challenge stable and hygienic anaerobic fermentation, and the system can be vulnerable to aerobic deterioration, fungal proliferation and nutrient loss if management is suboptimal [9,22]. These features make cattail-derived fermented forage matrices a relevant model for interrogating resistome dynamics under comparatively high-risk conditions. Meanwhile, shotgun metagenomics combined with ecological network inference and multivariate causal modeling enables joint profiling of ARGs, MGEs and microbial communities and helps identify key physicochemical and biotic drivers shaping mobile resistomes in complex ecosystems [23,24].

Here, we established a controlled cattail plant-biomass anaerobic fermentation system with and without inoculation of a locally selected *Lactiplantibacillus plantarum* and monitored fermentation characteristics, physicochemical properties, bacterial and fungal communities, ARGs and MGEs across anaerobic fermentation and a subsequent aerobic re-exposure phase using shotgun metagenomics. Co-occurrence network analysis, random forest models and partial least squares path modeling were employed to identify environmental variables, microbial groups and MGEs associated with resistome dynamics and to unravel potential pathways linking LAB inoculation to changes in ARGs and MGEs assemblages. We hypothesised that LAB inoculation would mitigate resistome-related risks by strengthening acidification and modifying organic acid profiles, thereby suppressing potential ARGs-host bacteria and reducing MGEs-associated mobility potential, with additional regulation arising from fungal–bacterial niche replacement during prolonged aerobic re-exposure.

2. Materials and methods

2.1. Materials preparation

Cattail samples were collected from the Hongyuan Experimental Base, affiliated with the Sichuan Academy of Grassland Sciences in Sichuan Province, China (32°46′N, 102°33′E, elevation 3500 m). The harvested cattail were shredded and chopped into 1–2 cm segments, then randomly allocated into four groups, each with three replicates. To reflect the routine use of lactic acid bacteria (LAB) inoculants in practical production and to compare the inoculation effects of LAB with different fermentation types, *Levilactobacillus brevis*(*L. brevis*), a heterofermentative LAB, and *Lactiplantibacillus plantarum*(*L. plantarum*), a predominantly homofermentative LAB, were selected for inoculation. The treatments were as follows: (1) CK, equal amounts of sterile water required for activation of bacteria are included; (2) *L. brevis* (LB, 10^6 cfu/g FM); (3) *L. plantarum* (LP, 10^6 cfu/g FM); and (4) LB + LP (M, 10^6 cfu/g FM). This comparative design was used to evaluate the effects of natural fermentation, single-strain inoculation, and mixed-strain inoculation strategies on fermentation characteristics, microbial succession, and ARGs dynamics in cattail silage during subsequent aerobic exposure. The CK group served as the uninoculated baseline for assessing natural fermentation and the natural succession of ARGs, whereas the LB, LP, and M groups were used to determine whether different LAB inoculation strategies produced distinct or complementary effects.

The inoculants used in this study included self-isolated *L. brevis* and *L. plantarum* strains. These strains were originally isolated from oat silage collected from the Qinghai–Tibetan Plateau and were selected from a laboratory lactic acid bacterial strain collection based on their acid-producing ability and growth performance during preliminary screening. They were identified as *L. brevis* and *L. plantarum* based on partial 16S rDNA sequencing, sequence similarity analysis, and preliminary physiological and biochemical characterization. Before application, the strains were activated and diluted with sterile water to obtain the target viable cell concentration of 10^6 CFU/g fresh matter. The inoculant suspension was sprayed evenly onto the chopped cattail using a hand sprayer, and the same volume of sterile water was added to the CK treatment. Additional phenotypic and preliminary biochemical characteristics of the two inoculant strains are provided in Supplementary Table S1. After inoculation and thorough mixing, approximately 2.0 kg of treated material was manually packed into polyethylene bags and vacuum-sealed for anaerobic fermentation for 60 days.

The aerobic exposure test followed the methodology described by [15]. In brief, the fermented cattail samples were placed in 1.0 L sterile plastic buckets, wrapped in two layers of coarse cotton fabric to reduce contamination and moisture loss, then kept at room temperature (25 °C). A thermocouple sensor was installed at the geometric center of the silage sample in each plastic container. These sensors were connected to an online 48-channel data logger (MDL-1048 A, Shanghai Tianhe Automation Instrumentation Co., Ltd., China), configured to record the temperature of each silage unit at 24-h intervals. Samples were collected at 3 d intervals during aerobic exposure, specifically on days 0, 3, 6, and 9. Day 0 represented the initial state before aerobic exposure, whereas days 3 and 6 were used to reflect stage-specific changes during aerobic exposure. Since previous studies have observed clear deterioration-related changes after approximately 7 d of aerobic exposure [25], the endpoint was set at 9 d to ensure that a clear late-stage aerobic deterioration state could be captured. A total of 48 samples (four inoculation treatments × four storage durations × three replicates) were included in the experimental design for subsequent analysis.

2.2. Sample composition analysis

A 10 g portion of each stored sample was homogenized with 90 mL of sterile water using a laboratory blender for 1 min, then filtered through four layers of gauze. The resulting filtrate was centrifuged at 4500 ×*g* for 15 min at 4 °C. Concentrations of butyric, propionic, acetic, and lactic acids were quantified via high-performance liquid chromatography (HPLC) as described by [26]. Ammoniacal nitrogen levels were measured following the protocol established by [27]. The pH of each sample was determined using a pH meter (APERA pH 700, Shanghai San-Xin Instrumentation Co., Shanghai, China).

Approximately 200 g of each storage sample was oven-dried at 65 °C until a constant weight was achieved to assess dry matter (DM) content,

then pulverized using a 0.20 mm mesh sieve for subsequent chemical analysis. Total nitrogen (TN) content was determined using the protocol of [28], and crude protein (CP) was estimated by multiplying TN by 6.25. The concentrations of neutral detergent fiber (NDF) and acid detergent fiber (ADF) were analyzed using the method proposed by [29]. Water-soluble carbohydrate (WSC) content was measured according to the procedure outlined by [30].

2.3. DNA extraction, metagenomic sequencing, and analysis of ARGs and MGEs

Genomic DNA from each sample was isolated using the CTAB extraction protocol. Subsequently, 1 μg of genomic DNA was randomly sheared into approximately 350 bp fragments using a Covaris ultrasonicator. Library construction included end repair, A-tailing, adaptor ligation, purification, and PCR amplification. Initial library quantification was performed using a Qubit 2.0 fluorometer, after which the library was diluted to 2 ng/μL. Insert size was evaluated using an Agilent 2100 Bioanalyzer. After confirming the expected insert size, the effective library concentration was quantified by qPCR, with a threshold of >3 nM used to validate library quality. Qualified libraries were pooled according to effective concentration and the intended sequencing data volume, followed by paired-end sequencing using the Illumina platform with a PE150 strategy.

Raw sequencing reads were preprocessed with Readfq (https://github.com/cjfields/readfq) to generate high-quality clean reads for downstream analyses. The filtering criteria included: (a) removal of reads containing a high proportion of low-quality bases (quality score $\leq$ 38, default threshold 40 bp); (b) removal of reads with excessive ambiguous bases (N; default threshold 10 bp); and (c) removal of reads with adapter overlap exceeding 15 bp. RawData represented the original sequencing data generated for each metagenomic sample, whereas CleanData represented the effective sequencing data retained after quality filtering. Only CleanData were used for subsequent assembly, gene prediction, taxonomic annotation, and ARGs/MGEs analyses.

Clean reads were assembled using MEGAHIT with the meta-large preset. Assembled scaffolds containing ambiguous bases (N) were split at N positions to generate scaftigs without Ns. Assembly quality was evaluated based on scaftigs longer than 500 bp, including the number of assembled scaftigs, average length, N50, and N90. These parameters were used to assess the continuity and quality of the metagenomic assemblies before downstream annotation.

Open reading frames were predicted from scaftigs longer than 500 bp using MetaGeneMark with default parameters. Predicted genes shorter than 100 nt were removed. CD-HIT was then used to remove redundant genes and construct a non-redundant unigene catalog, with the following parameters: -c 0.95, −G 0, -aS 0.9, −g 1, and -d 0. CleanData from each sample were mapped back to the non-redundant unigene catalog using Bowtie2 with the parameters –end-to-end, −-sensitive, —I 200, and -X 400. Genes with reads $\leq$2 in each sample were filtered out before downstream abundance analysis.

For abundance normalization, gene abundance was calculated based on the number of reads mapped to each unigene and the corresponding gene length. The normalized abundance of gene k was calculated as: $Gk = (rk/Lk) / \Sigma i(ri/Li)$. where Gk represents the normalized abundance of gene k, rk represents the number of reads mapped to gene k, and Lk represents the length of gene k. The normalized unigene abundance was then used to calculate taxonomic, ARGs, and MGEs relative abundances by summing the abundance of unigenes assigned to each corresponding category.

Unigene sequences were aligned against the NCBI NR database (https://www.ncbi.nlm.nih.gov/) using DIAMOND BLASTp (https://github.com/bbuchfink/diamond/) [31], with an *E*-value threshold of 1×10^{-5}. The analysis included Bacteria, Fungi, Archaea, and Viruses. For each sequence, alignments with E-values within 10 times the minimum E-value were retained. Because a single sequence can have multiple alignments, the lowest common ancestor (LCA) algorithm, as implemented in MEGAN, was used for taxonomic assignment [32]. The relative abundance of each taxon was calculated by summing the normalized abundance of unigenes assigned to that taxon.

Unigenes were aligned against the CARD database (https://card.mcmaster.ca/) using the Resistance Gene Identifier (RGI) tool provided by CARD. ARGs-like sequences were identified using the RGI built-in BLASTp algorithm with the default *E*-value threshold of <1 × 10–30. The relative abundance of each ARO, ARGs type, or ARGs subtype was calculated by integrating the normalized abundance of CARD-annotated unigenes assigned to the corresponding ARGs category [33–35].

For MGEs analysis, an ARGs-associated scaftig-based strategy was used. First, assembled scaftigs were compared with the CARD database to identify scaftigs carrying ARGs-like sequences. The corresponding ARGs-containing scaftigs were then extracted according to CARD annotation results and gene prediction files. These ARGs-containing scaftigs were further compared with MGEs databases, including Integrall (version 2021.09), ISfinder (version 2023.10), and Plasmid (version 2021.06), to screen for integrase-, insertion sequence-, and plasmid-associated elements located on ARGs-associated scaftigs. BLASTn hits with sequence identity $\geq$90% and *E*-value $\leq 1 \times 10^{-7}$ were retained as MGEs-like sequences. These thresholds were used to reduce weak or ambiguous matches and improve the stringency of MGEs annotation. The ARGs and MGEs mining results from scaftigs were summarized for downstream analyses of ARGs-associated mobility potential.

2.4. Statistical analysis

A general linear model (GLM) was employed using the default GLM procedure in SPSS software (IBM Corp., Armonk, NY, USA) to evaluate the individual and interactive effects of LAB inoculant treatment and aerobic exposure duration on the fermentation characteristics and chemical composition of cattail silage during aerobic exposure. Duncan's multiple range test was used for mean separation when significant effects were detected, and statistical significance was declared at $P < 0.05$. Multiple comparisons of ARGs and MGEs abundances among LAB inoculant treatments and aerobic exposure durations were performed separately using GraphPad Prism.

Co-occurrence network analysis was conducted to explore potential associations among ARGs, MGEs, and microbial taxa. Microbial taxa with detection frequencies below 60% and relative abundances below 0.5% were removed before network construction. Spearman's correlations were calculated among ARGs, MGEs, and retained microbial taxa, and *P* values were adjusted using the Benjamini–Hochberg procedure. Only strong and significant correlations with $|R| > 0.8$ and adjusted $P <$ 0.05 were retained. Networks were constructed in R and visualized using Gephi version 0.10. Network relationships were interpreted as ecological associations rather than direct evidence of host-resolved ARGs transfer.

Random forest modeling, Mantel tests, Procrustes analysis, and partial least squares path modeling (PLS-PM) were conducted in R. Random forest models were used to identify the major environmental, chemical, and microbial predictors of ARGs abundance. Model performance was evaluated using the out-of-bag prediction procedure, and variable importance was quantified using the increase in mean squared error (%IncMSE). The stability of variable importance was assessed using bootstrap-derived standard errors and 95% confidence intervals, and Boruta analysis was applied to classify predictors as confirmed, tentative, or rejected variables.

Mantel tests and Procrustes analysis were performed using the vegan package to evaluate the relationships among ARGs profiles, MGEs profiles, microbial communities, and environmental variables. Bray–Curtis dissimilarity was used for ARGs profiles, MGEs profiles, and microbial communities, whereas Euclidean distance was used for standardized environmental and chemical variables. Mantel correlations were

calculated using Spearman's method, and the significance of Mantel and Procrustes statistics was assessed using permutation tests.

PLS-PM was performed using the plspm package to evaluate putative association pathways linking environmental factors, chemical composition, microbial communities, MGEs, and ARGs. Variables were log-transformed when necessary and standardized before analysis. Model performance was evaluated based on path coefficients, R^2 values of endogenous latent variables, and the global goodness-of-fit index. The robustness of the association-based conclusions was further evaluated by comparing the directional consistency among random forest modeling, Mantel tests, Procrustes analysis, co-occurrence network analysis, and PLS-PM. Therefore, the identified pathways and network relationships were interpreted as association-based ecological patterns rather than definitive causal relationships.

3. Results and discussion

3.1. Succession of ARGs and MGEs under aerobic exposure

To contextualize resistome succession during aerobic exposure, we first characterized the fermentation microenvironment and substrate matrix of cattail silage. Across 0–9 d exposure, pH increased and ammonia-N accumulated at the late stage, while fermentation acids were progressively depleted, with the lowest lactic acid and acetic acid levels observed at mid-exposure (Table 1). In parallel, dry matter increased and structural carbohydrate fractions (NDF and ADF) became enriched, whereas WSC declined during mid-exposure (Table 2), consistent with substrate depletion and compositional concentration effects under oxygen ingress (Peng et al., 2018). Notably, LP established the strongest initial acidification and maintained lower ammonia-N, providing a distinct physicochemical niche that can shape microbial succession and downstream ARGs and MGEs dynamics. Against this physicochemical backdrop, we next characterized the dynamics of ARGs and MGEs across treatments and exposure duration.

A total of 23 ARGs types and 171 ARGs subtypes were identified across 48 silage samples (Fig. 1a and Table 3). Among these ARGs, 10 types were consistently detected (100% of the ARGs types), with glycopeptide antibiotic and tetracycline antibiotic being the most abundant (Fig. 1a and Table 3). Glycopeptide antibiotic and tetracycline antibiotic evidently underpinned the variation in total ARGs abundance (Fig. 1b). Among the ARGs subtypes, tetM had the highest abundance, followed by the vanY gene in the vanG cluster and the vanY gene in the vanM cluster. Zhao et al. demonstrated that the distribution of ARGs is related to ARGs carried by different bacteria [36], and different lactic acid bacteria lead to changes in ARGs abundance during the silage process [18]. A similar phenomenon was observed in this study: ARGs distributions differed among treatments with various additives and shifted progressively with exposure duration (Fig. 1b). Overall, the reduction effect on ARGs was achieved by the 9th day of aerobic exposure. Surprisingly, the CK treatment exhibited a trend of increasing followed by decreasing total ARGs abundance, whereas LP treatment markedly reduced overall ARGs levels. This is consistent with the observation by reference Zhang et al. that LP additives reduced ARGs in alfalfa silage [12]. Exogenous lactic acid bacterial additives frequently harbor a small number of ARGs [20], contributing to elevated ARGs risks in silage feed. LB treatment possibly contained tetracycline ARGs, enhancing tetracycline antibiotic abundance (Fig. 1b) and consequently raising its overall abundance. Mixed additives often lead to competition among bacterial strains, affecting the microbial community [37]. This is the fundamental reason for the changes in ARGs under the M treatment. Variations in ARGs abundance in the M treatment suggest that the mixed additives in this study failed to eliminate ARGs.

Variance analysis was used to characterize the changes in total ARGs abundance under different treatments (interaction between exposure

Table 1
Dynamics of fermentation characteristics in cattail silage during aerobic exposure.

Items	Treatment (T)	Aerobic exposure (D)					SEM	*P*-value		
		day 0	day 3	day 6	day 9	Mean^{T}		T	D	T × D
pH	CK	4.35	4.44	4.40	5.13	4.58^{A}	0.065	0.114	0.001	0.672
	LB	4.13	4.15	4.32	4.57	4.29^{AB}				
	LP	3.98	4.10	4.62	5.08	4.45^{AB}				
	M	4.14	4.13	4.20	4.38	4.21^{B}				
	Mean^{D}	4.15^{b}	4.20^{b}	4.39^{b}	4.79^{a}					
Ammonia-N, %TN	CK	1.61	1.73	1.27	2.84	1.86^{A}	0.117	0.016	<0.001	0.396
	LB	1.81	2.20	1.04	2.89	1.99^{A}				
	LP	1.35	1.01	0.96	2.20	1.38^{B}				
	M	2.26	1.41	1.03	3.19	1.97^{A}				
	Mean^{D}	1.76^{b}	1.59^{b}	1.08^{c}	2.78^{a}					
Lactic acid, %DM	CK	2.69^{B}	1.39^{B}	1.15	1.72	1.74	0.157	<0.001	<0.001	0.002
	LB	2.83^{B}	3.93^{A}	1.56	2.87	2.80				
	LP	4.18^{Aa}	3.68^{Aa}	0.90^{c}	2.36^{b}	2.78				
	M	2.55^{Ba}	1.75^{Bbc}	1.36^{c}	2.38^{ab}	2.01				
	Mean^{D}	3.06	2.69	1.24	2.33					
Acetic acid, %DM	CK	0.23	0.19	0.15	0.10	0.17^{A}	0.012	0.007	0.02	0.781
	LB	0.16	0.14	0.16	0.05	0.13^{AB}				
	LP	0.13	0.08	0.02	0.03	0.06^{C}				
	M	0.14	0.08	0.08	0.10	0.10^{BC}				
	Mean^{D}	0.16^{a}	0.13^{ab}	0.10^{b}	0.07^{b}					
Propionic acid, %DM	CK	0.96	0.78	0.50	0.85	0.77^{A}	0.043	0.005	0.01	0.938
	LB	0.72	0.94	0.57	0.91	0.78^{A}				
	LP	0.59	0.49	0.26	0.40	0.43^{B}				
	M	0.78	0.84	0.46	0.86	0.74^{A}				
	Mean^{D}	0.76^{a}	0.76^{a}	0.45^{b}	0.76^{a}					
Butyric acid, %DM	CK	0.07	0.07	0.05	0.10	0.07^{A}	0.004	0.057	0.005	0.771
	LB	0.04	0.05	0.04	0.05	0.05^{B}				
	LP	0.04	0.05	0.04	0.09	0.06^{AB}				
	M	0.06	0.06	0.05	0.07	0.06^{AB}				
	Mean^{D}	0.05^{b}	0.06^{b}	0.04^{b}	0.08^{a}					

DM, dry matter; SEM, standard error of the total mean; Mean ^{T}, Overall mean of treatments; Mean ^{D}, Overall mean of aerobic exposure days; CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*. Means within the same row (a–e) or within the same column (A–D) with different superscripts differ significantly from each other ($P < 0.05$).

Table 2
Dynamics of chemical composition of cattail silage during aerobic exposure.

Items	Treatment (T)	Aerobic exposure (D)					SEM	*P*-value		
		day0	day3	day6	day9	Mean^{T}		T	D	T × D
DM,% FM	CK	20.36	22.21	21.52	25.29	22.34^{B}	0.35	0.009	<0.001	0.984
	LB	20.71	23.93	21.58	24.43	22.66^{B}				
	LP	22.13	25.22	24.44	26.90	24.67 ^{A}				
	M	20.23	22.75	21.93	24.34	22.31^{B}				
	Mean^{D}	20.86^{c}	23.53^{b}	22.37^{bc}	25.24^{a}					
Water soluble carbohydrates, %DM	CK	1.10^{BCa}	0.71^{b}	0.88^{ab}	0.87^{Bab}	0.89	0.072	<0.001	<0.001	0.033
	LB	1.47^{ABa}	0.87^{bc}	0.36^{c}	1.13^{Bab}	0.96				
	LP	1.79^{Aab}	1.02^{b}	1.02^{b}	2.18^{Aa}	1.50				
	M	0.96^{C}	0.91	1.03	1.22^{B}	1.03				
	Mean^{D}	1.33	0.88	0.82	1.35					
Crude protein, %DM	CK	12.28	13.85	12.91	12.12	12.79^{A}	0.141	0.031	<0.001	0.102
	LB	11.29	13.29	11.88	11.32	11.95^{B}				
	LP	13.12	13.28	11.59	11.18	12.29^{AB}				
	M	12.08	13.11	12.81	12.03	12.51^{AB}				
	Mean^{D}	12.19^{bc}	13.39^{a}	12.30^{b}	11.66^{c}					
Neutral detergent fiber, %DM	CK	55.56	53.00	57.83	60.86	56.82	0.722	0.188	<0.001	0.142
	LB	53.72	51.94	53.39	63.29	55.58				
	LP	53.36	52.17	61.95	64.10	57.90				
	M	56.18	50.94	56.18	56.19	54.87				
	Mean^{D}	54.7^{bc}	52.01^{c}	57.34^{b}	61.11^{a}					
Acid detergent fiber, %DM	CK	29.47	26.05	29.54	31.99	29.26	0.48	0.263	<0.001	0.003
	LB	27.44	24.52	32.36	33.23	29.39				
	LP	25.59	26.32	33.27	33.85	29.76				
	M	29.70	25.27	25.27	29.36	28.35				
	Mean^{D}	28.05^{b}	25.54^{c}	31.06^{a}	32.10^{a}					

DM, dry matter; FM, fresh matter; SEM, standard error of the total mean; Mean ^{T}, Overall mean of treatments; Mean ^{D}, Overall mean of aerobic exposure days; CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*. Means within the same row (a–e) or within the same column (A–D) with different superscripts differ significantly from each other ($P < 0.05$).

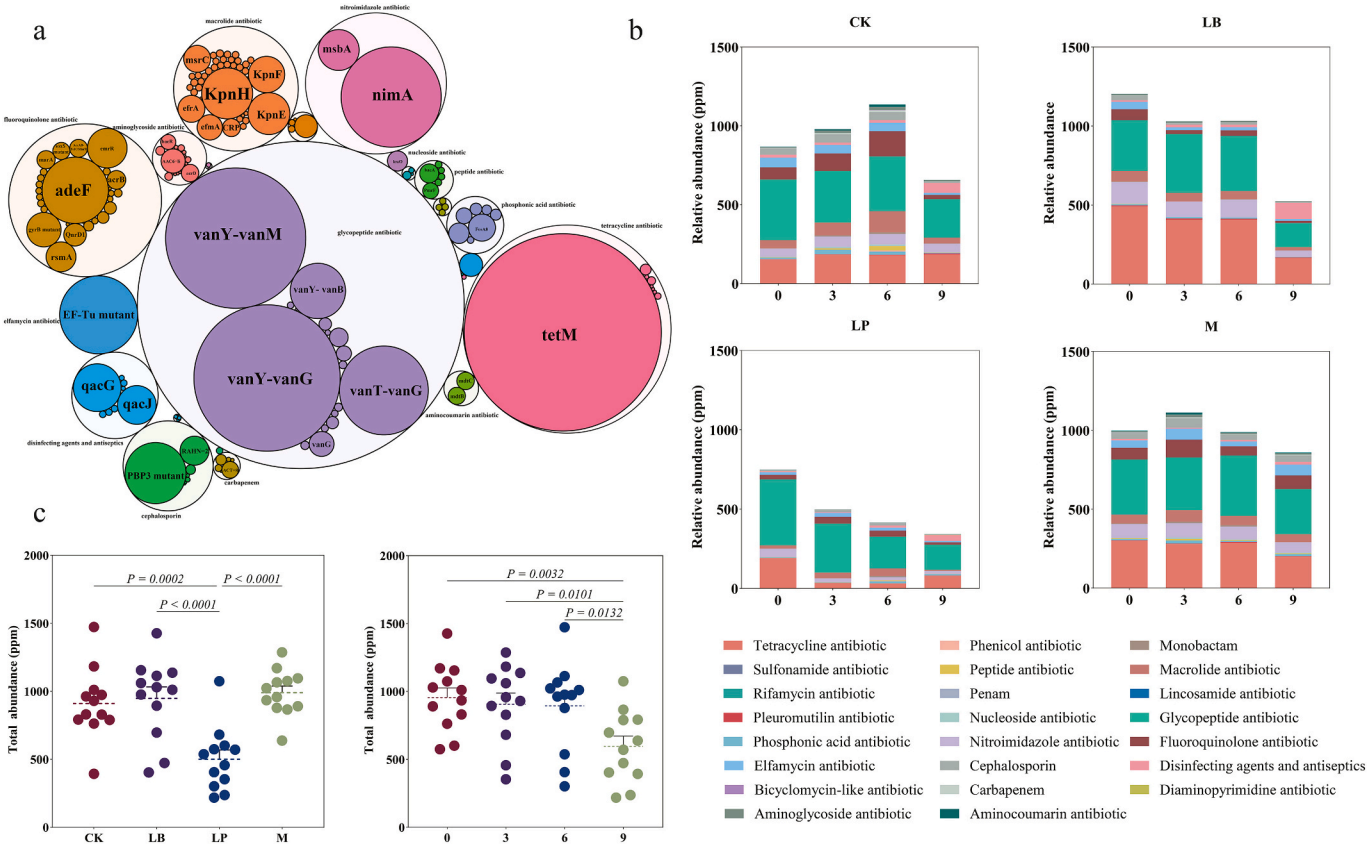


Fig. 1. Composition and distribution of cattail silage ARGs. (a) ARGs compositions are colored according to the ARGs type. The outer and inner circles represent ARG types and subtypes, respectively. Circle size is proportional to the relative abundance of ARGs; (b) Relative abundance of ARGs during aerobic exposure of cattail; (c) Distribution of ARGs total relative abundance under different treatments(left) and across exposure days (right). CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of L. *brevis* and L. *plantarum*.

Table 3
Number and relative abundance of ARG types.

ARG types	Number of ARG subtypes	Detected count	Detection frequency	Minimal abundance (ppm)	Maximal abundance (ppm)	Average abundance (ppm)	Percentage of total mean abundance
aminocoumarin antibiotic	2	46	95.83%	0.00	41.62	3.43	0.41
aminoglycoside antibiotic	16	48	100.00%	0.10	39.50	7.99	0.95
bicyclomycin-like antibiotic	1	7	14.58%	0.00	2.56	0.24	0.03
carbapenem	6	46	95.83%	0.00	27.98	2.69	0.32
cephalosporin	7	48	100.00%	0.69	90.67	26.86	3.21
diaminopyrimidine antibiotic	5	15	31.25%	0.00	4.09	0.21	0.03
disinfecting agents and antiseptics	8	48	100.00%	1.13	239.34	22.80	2.72
elfamycin antibiotic	1	48	100.00%	5.84	126.12	34.73	4.15
fluoroquinolone antibiotic	28	48	100.00%	10.61	295.96	61.45	7.34
glycopeptide antibiotic	17	48	100.00%	78.89	462.95	307.97	36.77
lincosamide antibiotic	2	27	56.25%	0.00	1.26	0.13	0.02
macrolide antibiotic	39	48	100.00%	5.65	188.02	53.21	6.35
monobactam	4	44	91.67%	0.00	25.23	3.48	0.42
nitroimidazole antibiotic	2	48	100.00%	3.08	166.89	66.84	7.98
nucleoside antibiotic	1	43	89.58%	0.00	26.12	2.28	0.27
penam	3	43	89.58%	0.00	5.00	0.58	0.07
peptide antibiotic	6	46	95.83%	0.00	63.84	4.46	0.53
phenicol antibiotic	5	14	29.17%	0.00	7.77	0.92	0.11
phosphonic acid antibiotic	7	48	100.00%	0.96	49.80	10.26	1.22
pleuromutilin antibiotic	1	47	97.92%	0.00	17.85	3.04	0.36
rifamycin antibiotic	1	4	8.33%	0.00	1.26	0.03	0.00
sulfonamide antibiotic	1	7	14.58%	0.00	4.74	0.11	0.01
tetracycline antibiotic	8	48	100.00%	7.05	567.83	223.89	26.73
Total	171	48	100.00%	114.00	2456.40	837.60	100.00

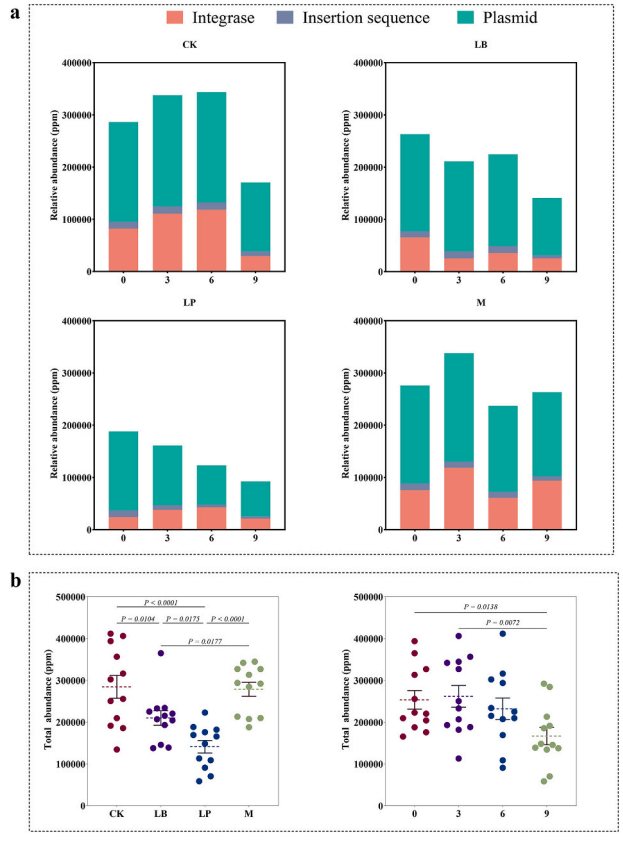


Fig. 2. Composition and distribution of cattail silage MGEs. (a) Relative abundance of MGEs during aerobic exposure of cattail; (b) Distribution of total relative abundance of MGEs across treatments and time. CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*.

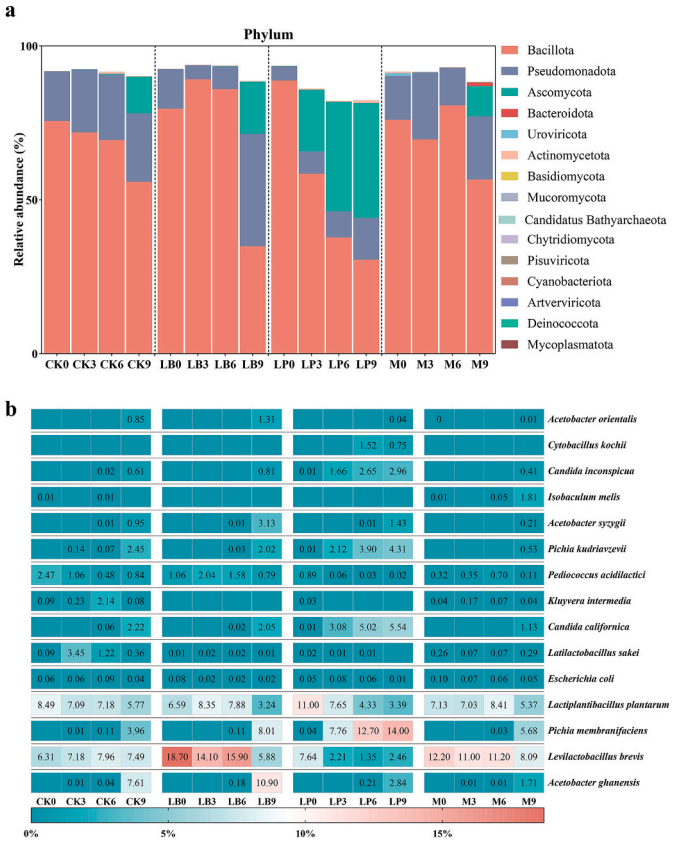


Fig. 3. Relative abundance of bacterial communities at the (a) phylum level and (b) species level (top 15) during aerobic exposure of cattail. CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*.

days and treatment was not significant, Fig. 1b, c), revealing differences in ARGs abundance among treatments. As a homofermentative lactic acid bacterium, *Lactiplantibacillus plantarum* efficiently produces lactic acid, lowering pH and indirectly suppressing the proliferation of ARGs-hosting bacteria [38]. Interestingly, the relative abundance of ARGs decreased after prolonged exposure. In the early stage of aerobic exposure, the increase in oxygen and nutrient availability can stimulate the growth of potential ARGs hosts such as *Enterobacteriaceae* [10,39], leading to a transient rise in ARGs abundance. However, when exposure is extended, oxygen accumulation and substrate depletion cause oxidative stress and oxygen toxicity, which inhibit the survival of facultative and obligate anaerobic ARGs hosts [40]. Similarly, Wang et al. also observed a decrease in ARGs abundance over time during aerobic composting fermentation, attributing this to fermentation environment-induced changes in ARGs hosts [41]. Therefore, our results suggest a time-dependent response in ARGs dynamics: early aerobic exposure activates ARGs-hosting bacteria, while prolonged exposure induces oxygen toxicity and environmentally mediated bacterial community restructuring, suppressing ARGs hosts and lowering ARGs abundance. This also explains the initial increase followed by a decline in ARGs abundance in the CK treatment, and the treatment-specific differences further corroborate the community restructuring hypothesis. Although extended exposure reduces ARGs risk, it also increases silage spoilage, implying that natural aerobic deterioration during feeding still poses potential ARGs hazards (Fig. 1b).

The dissemination of ARGs is closely related to MGEs. To assess their transmission risk, we annotated three major MGEs (Fig. 2). Across all treatments, the variations in MGEs were predominantly influenced by Integrase and Plasmid, which represented the majority share. Thus, Integrase and Plasmid emerge as the principal contributors to the overall dynamics of MGEs. Moreover, during aerobic exposure, changes in MGEs across all treatments exhibited a synchronous trend with ARGs abundance, further highlighting the critical role of MGEs in driving ARGs risks. Elevated MGE abundance signals a substantial increase in the risk of horizontal ARGs transfer [42]. Multiple comparison analysis of total MGEs revealed (no significant interaction between exposure days and treatments, Fig. 2b, c) that LP and LB treatments significantly reduced the overall MGEs abundance, but their combination exhibited antagonistic effects. This is probably because competition among microorganisms affects the changes in MGEs. Regarding the duration of aerobic exposure, the overall abundance of MGEs gradually decreased with increasing exposure days. MGEs are also carried by microbes, and these changes are consistent with our previous discussion on microbial competition and the influence of environmental conditions on microbial dynamics. Specifically, single-strain treatments were significantly more effective than multi-strain treatments, and environmental changes had a notable impact on the succession patterns of MGEs.

3.2. Microbial community succession during aerobic exposure

Microbes act as carriers of ARGs, and their succession indicates changes in ARGs. Annotating microbial communities helps to analyze ARGs risk (Fig. 3). During all aerobic exposures, Bacillota and Pseudomonadota were dominant. As aerobic exposure increased, the relative abundance of Bacillota gradually decreased, a result attributed to the proliferation of aerobic bacteria limiting Bacillota activity [43] Bacillota are major producers of ARGs, responsible for carrying and disseminating ARGs [44]. The changes in Bacillota were consistent with the changes in ARGs, explaining the observed reduction in ARGs in this study. Moreover, Pseudomonadota comprises many aerobic pathogens, including *Enterobacteriaceae,* potential hosts for ARGs [45]. Aerobic conditions promoted the rise of Pseudomonadota, resulting in increased ARGs in CK treatment between days 3–6. Overall, the duration of aerobic exposure influenced the silage bacterial community, and the succession of Bacillota and Pseudomonadota significantly impacted ARGs. Notably, Ascomycota showed high abundance in the late stages of aerobic exposure, suggesting that it may influence the silage environment and determine the fate of ARGs during fermentation. Ascomycota is often recognized as a key fungal group for breaking down lactic acid in silage (e.g., yeast). Some yeasts utilize lactic acid as a substrate [46], which may explain their increase relative to the initial lactic acid (Table 1). Traditionally, the low pH environment caused by high lactic acid inhibits the proliferation of most Pseudomonadota, including undesirable Enterobacteriaceae in silage [16]. At day 0, the lowest pH in the LP treatment inhibited the growth of most potential ARGs hosts (Table 1), resulting in the lowest ARGs abundance. However, excessive lactic acid supported the proliferation of Ascomycota in LP. Further investigation is needed to determine if Ascomycota amplification involves potential ARGs carriers and its specific role in ARGs dissemination risks. Therefore, analysis of microbial communities at the host species level will help reveal the potential roles and mechanisms of Ascomycota in ARGs dissemination risks.

At the species level, *Lactiplantibacillus plantarum* (*L. plantarum*) and *Levilactobacillus brevis* (*L. brevis*) are the dominant lactic acid bacteria during fermentation, with their high abundance closely linked to lactic acid production and maintenance of a low pH environment, significantly inhibiting the growth of potential pathogens (e.g., *Escherichia coli*) and reducing ARGs dissemination risk [47]. However, the potential role of L. *brevis* as a host of Tetracycline ARGs cannot be overlooked. The changes in L. *brevis* under M and LB treatments were consistent with changes in Tetracycline relative abundance, supporting our earlier hypothesis. Roberts and Schwarz reviewed the distribution of resistance genes, including Tetracycline, among agricultural and environmental bacteria, clearly highlighting the widespread presence of tetM, tetO, and other Tetracycline resistance genes in lactic acid bacteria such as L. *brevis*, emphasizing their potential role as a resistance gene reservoir [48]. Evidently, when considering its use as a silage fermentation promoter, the potential risk of ARGs dissemination from L. *brevis* must be thoroughly evaluated. On the other hand, in the later stages of aerobic exposure, the abundance of yeasts such as *Pichia membranifaciens, Pichia kudriavzevii,* and *Candida californica* increased significantly. These yeasts may indirectly regulate microbial community structure by metabolizing lactic acid, altering environmental pH and chemical composition [49]. This regulatory effect might suppress the proliferation of Bacillota bacteria, the primary carriers of ARGs (Fig. 3a), ultimately reducing ARGs abundance and dissemination risk. Furthermore, Li et al. highlighted in composting studies that microbial consortia, including yeasts, significantly enhanced ARGs removal rates by raising compost temperatures, accelerating water loss, and promoting nitrogen (ammoniacal nitrogen) fixation [50]. Similarly, Cao et al. found that microbial compost inoculants with yeasts reduced ARGs abundance, and this reduction correlated with a decrease in Bacillota abundance [51]. Although yeast proliferation might limit the growth of ARGs hosts, explaining the reduction in ARGs abundance during LP treatment and late aerobic exposure, further studies are needed to determine whether they harbor potential ARGs carriers. Moreover, the specific mechanisms of yeast involvement in ARGs dissemination risks remain unclear, and their potential role in horizontal gene transfer needs to be explored further.

3.3. Detection of potential ARGs hosts in cattail silage under aerobic exposure

ARGs host identification at the phylum level was conducted using metagenomics (Fig. 4a). ARGs hosts were predominantly identified as Bacillota and Pseudomonadota, supporting our previous assumptions. Nonetheless, metagenomic identification is limited [23], leaving some potential microbial species unidentified. Therefore, the co-occurrence of ARGs, MGEs, and microbial communities was used to identify potential ARGs host microbes [52]. The results showed similarity between LB and M, with LB generating 131 nodes and 176 edges, while M had 118 nodes and 164 edges. Despite differences in node and edge counts, positive correlations between ARGs and microbial communities dominated both LB and M treatments, suggesting that ARGs dynamics were closely

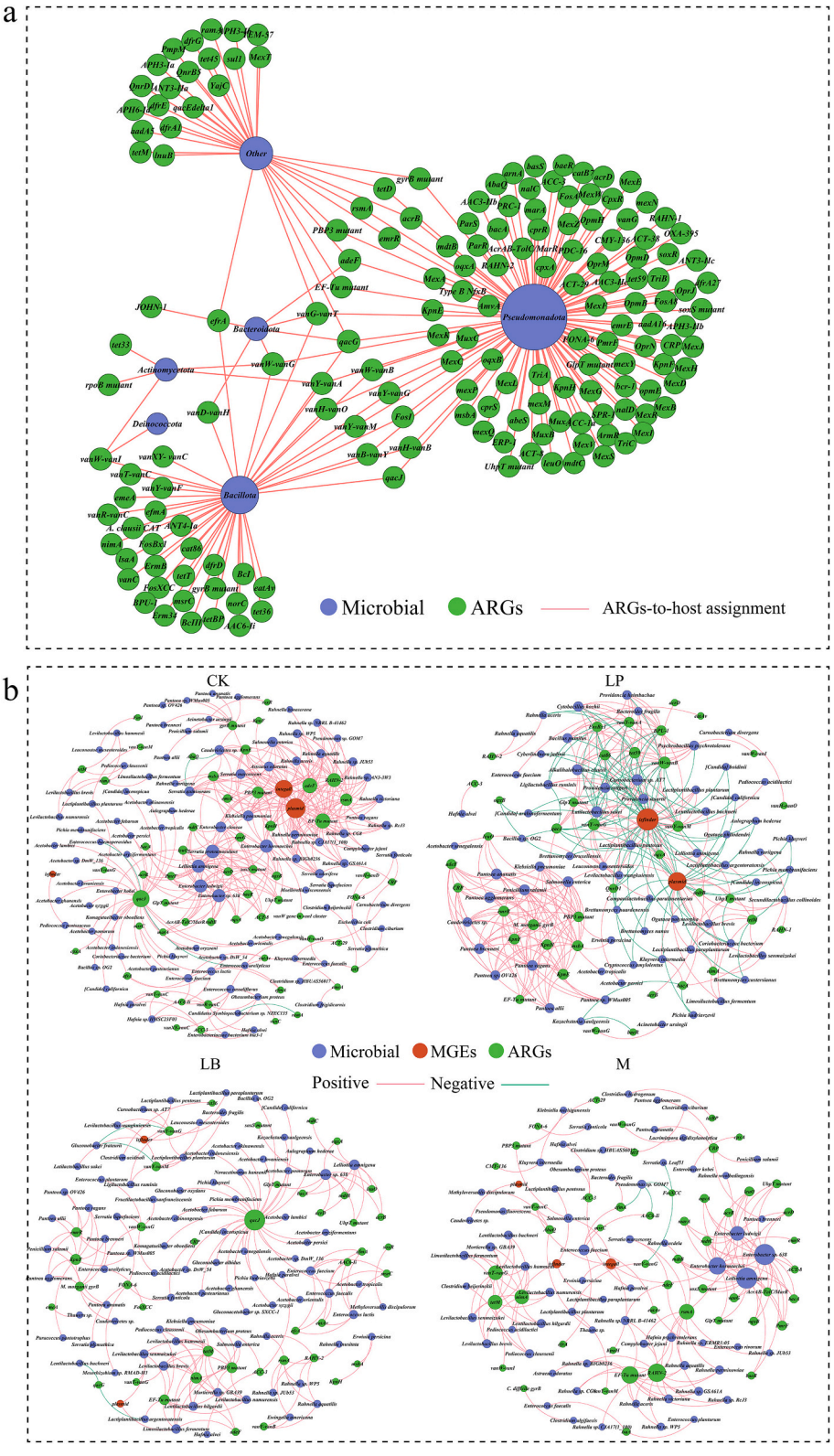


Fig. 4. Metagenome-based host assignment and co-occurrence patterns of ARGs during aerobic exposure of cattail silage. (a) Host assignment of ARGs based on metagenomic annotations. Red lines indicate the assignment of ARGs to their corresponding microbial hosts. (b) Co-occurrence network among ARGs, MGEs, and potential host bacteria during aerobic exposure. CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*.

associated with bacterial proliferation and interactions with MGEs. The qacJ ARGs emerged as notable key nodes in the LB treatment, being carried by various bacterial communities. Their high abundance may have enhanced connectivity between ARGs and host bacteria. Comparatively, the network structure in the M treatment had slightly fewer nodes and edges, but the associations among ARGs, MGEs, and microbial communities remained tight. Uniquely, specific bacteria such as *Enterobacter hormaechei* and *Erwinia persicina* in the M treatment were associated with integrase-related MGEs, which may have increased ARGs dissemination potential and contributed to the higher ARGs abundance observed in the late aerobic exposure stage (Fig. 1b).

For CK and LP, 155 nodes with 358 connections and 108 nodes with 322 connections were generated, respectively. The network shows that most microbes in CK carried ARGs and were closely associated with MGEs, and such multiple interactions significantly increased the risk of ARGs dissemination. This pattern is consistent with previous evidence that horizontal gene transfer can be an important determinant of ARGs profiles [53], suggesting that, under conditions without inoculated additives, MGEs-associated mobility potential and microbial proliferation jointly contributed to ARGs dissemination risk. Especially certain highly connected nodes, such as *Enterobacteriaceae* and other potential pathogens, may play central roles in ARGs dissemination networks, significantly enhancing the risk of ARGs spread. Conversely, qacJ ARGs in CK were like those in LB treatment, as they were driven by diverse bacterial hosts under aerobic exposure, resulting in elevated ARGs levels. The multi-host mechanism maintained stable ARGs growth [36], explaining the notable ARGs increase observed in the early aerobic exposure of CK treatment. In contrast, LP treatment significantly simplified the network structure, reducing positive correlations among ARGs, MGEs, and microbes while increasing negative correlations, effectively lowering ARGs dissemination risk. The suppression is largely due to the low pH environment driven by the dominance of L. *plantarum.* The low pH not only inhibited the proliferation of ARGs host bacteria but also disrupted MGEs activity, weakening ARGs dissemination within the microbial community. Studies have shown that low pH environments can significantly inhibit the metabolic activity of sensitive microbes and reduce their gene transfer capability [54], lowering the likelihood of ARGs dissemination via MGEs. This result further supports the important role of L. *plantarum* in reducing ARGs-related risks and highlights a possible pathway by which a low-pH environment reshapes microbial communities and limits ARGs dissemination potential. Additionally, in all four treatments, *L. brevis* was significantly positively correlated with tetM, indicating its potential role as a Tetracycline host and supporting our earlier inference that adding L. *brevis* increased ARGs abundance.

However, the stronger ARGs attenuation observed in the LP treatment should not be attributed solely to a strain-specific function. LP established a stronger acidified environment and reduced ammonia-N accumulation, which may have created unfavorable conditions for potential ARGs-hosting bacteria and reduced MGEs-associated mobility potential. Therefore, the LP-associated decrease in ARGs abundance may reflect both the inoculation effect of the self-isolated L. *plantarum* strain and the broader improvement in fermentation quality caused by stronger acidification and reduced protein degradation. Because acidification-matched controls were not included and L. *plantarum* strains with different acidification capacities were not compared in this study, it remains difficult to fully separate the strain-specific effect of LP from fermentation-quality-driven effects. Future studies should incorporate acidification-matched controls and multiple L. *plantarum* strains with contrasting acidification capacities to further distinguish strain-specific effects from fermentation-quality-driven effects.

3.4. Contributions of MGEs, microbes, and silage environment to ARGs dissemination in aerobic exposure of cattail silage

This study explored the relationships between ARGs, microbial communities, MGEs, and environmental factors using Procrustes analysis and Mantel tests (Fig. 5). The results showed a significant correlation between the spatial distribution of ARGs and MGEs ($M^2 = 0.33$, $P = 0.001$), indicating that MGEs may play an important role in ARGs mobility potential. Moreover, ARGs exhibited a strong association with microbial communities ($M^2 = 0.41$, $P = 0.001$), suggesting a possible link between bacterial population changes and ARGs dissemination potential. Nonetheless, Mantel test analysis also revealed a strong correlation between ARGs and microbial communities ($R = 0.833$, $P = 0.001$), indicating that ARGs variation was not solely related to MGEs-associated mobility but may also be influenced by microbial community structure. Furthermore, environmental factors also played a significant role in ARGs variation ($R = 0.289$, $P = 0.001$). They may indirectly influence MGEs-associated mobility potential through selection pressure, thereby affecting ARGs dynamics. In conclusion, the evidence from this study suggests that MGEs may contribute importantly to ARGs dissemination potential, while microbial community structure and environmental factors also play regulatory roles. However, because direct transfer events and host-resolved ARGs and MGEs linkages were not experimentally validated, these findings should be interpreted as ecological associations and mechanistic indications rather than definitive evidence that horizontal gene transfer was the dominant route.

To determine the impact of environmental and microbial factors on ARGs during aerobic exposure, random partial dependence plots were constructed using randomForest, rfUtilities, rfPermute, and tidyverse packages to quantify the response of key environmental factors and microbes to ARGs dissemination (Fig. 6a, b). The results showed that ARGs were primarily influenced by microbes such as L. *brevis, L. plantarum,* and *Pichia membranifaciens*. This aligns with our earlier inference that different lactic acid bacteria inoculants and the increased yeast during exposure are the primary microbial drivers of ARGs dissemination. Regarding environmental factors, DM, NDF, acetic acid, pH, temperature and ADF were dominant. Environmental factors significantly influenced the distribution and abundance of ARGs by regulating microbial com Specifically, moisture content (DM), temperature, acidic conditions (acetic acid, pH), and substrate availability (fiber) acted on potential ARGs host bacteria, driving the dynamic changes in ARGs [8,24,50,55]. Consequently, the partial least squares path model (PLS-PM) highlighted the direct and indirect impacts of environmental factors (temperature, acetic acid, pH), chemical components (DM, NDF, ADF), and key microbial communities (top 5 in importance) on ARGs variations (Fig. 6c). The results showed that environmental factors indirectly drove ARGs changes by directly regulating microbial communities ($\lambda = 0.378$, $P < 0.05$). Moreover, microbial communities not only directly influenced ARGs abundance ($\lambda = 0.429$, $P < 0.01$) but also significantly affected MGEs profiles ($\lambda = 0.731$, $P < 0.01$), further influencing ARGs dissemination potential. Chemical composition has no direct effect on ARGs ($P > 0.05$) but primarily influences the spread of ARGs by regulating microbial communities ($\lambda = 0.418$, $P < 0.01$).

The PLS-PM model further highlighted the important role of MGEs in ARGs variation ($\lambda = 0.484$, $P < 0.01$). This result indicates that, within complex microbial communities, ARGs dynamics are closely related to MGEs profiles, which may be jointly influenced by environmental and chemical factors. However, this regulation occurs primarily indirectly through microbial mediation, as neither environmental nor chemical components exhibited significant direct effects on MGEs ($P > 0.05$). Instead, both strongly shaped microbial community structure ($P < 0.05$). The significant correlations among yeast, LAB, and MGEs further supported this microbially associated pattern (Fig. 6c). This finding is consistent with previous studies suggesting that environmental matrices influence ARGs behavior by reshaping microbial community structure [56]. Notably, the effects of environmental and chemical factors on ARGs largely depend on changes in microbial community structure, while the microbial regulation of ARGs is mainly mediated by MGEs. This suggests that microbial community responses remain a key driver of ARGs variation during aerobic exposure. The application of different lactic acid bacteria inoculants significantly altered microbial

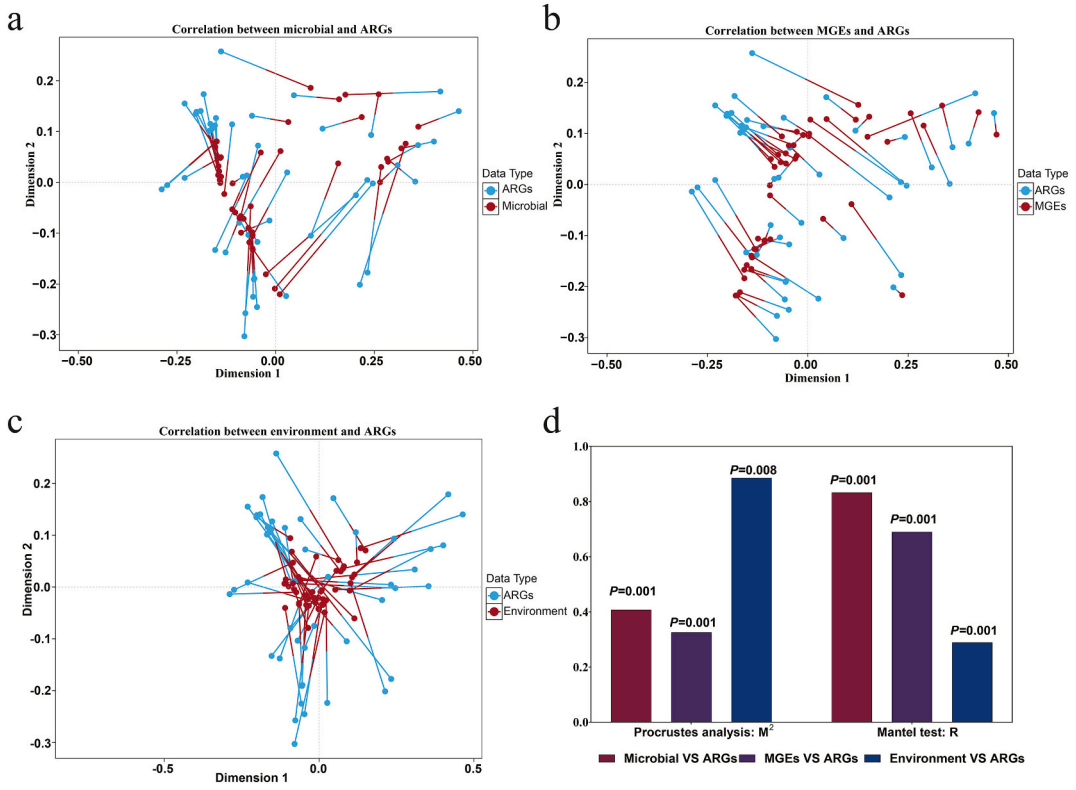


Fig. 5. Relationships between ARGs and microbial communities, MGEs and environmental factors during aerobic exposure. Procrustes analysis shows the concordance of ARGs with microbial communities (a), MGEs (b) and environmental factors (c). (d) Procrustes M^2 and Mantel R values summarize the associations between ARGs and the corresponding factors.

composition and ecological functions, thereby influencing ARGs dynamics. Specifically, dominant LAB species such as *Lactiplantibacillus plantarum* and *Levilactobacillus brevis* optimized the fermentation environment by lowering pH and accumulating acetic acid, which suppressed the proliferation of conditional pathogens and MGEs. In the later stage of exposure, the proliferation of certain yeasts such as *Pichia*

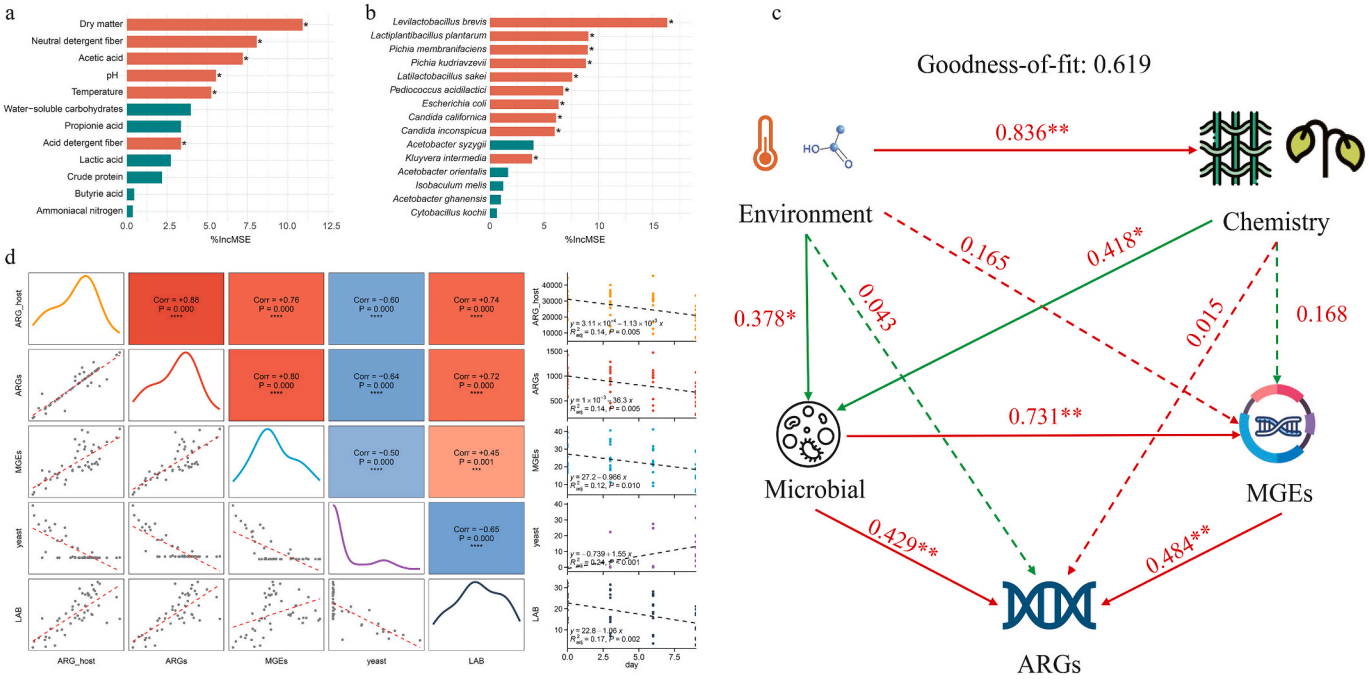


Fig. 6. Environmental factors modulate the spread of ARGs through microbial. Random forest distribution maps reveal important environmental factors (a) and microbial (b) affecting ARGs; (c) PLS-PM shows the direct and indirect effects of key factors on the amplification of ARGs. (d) The correlation analysis between Yeast, LAB, ARGs, MGEs and ARGs host. Red and green arrows indicate positive and negative effects, respectively. Dashed arrows indicate nonsignificant. *, $P < 0.05$, **, $P < 0.01$.

membranifaciens further intensified substrate competition and altered acidity and substrate availability, thereby reducing the growth advantage of ARGs-harboring bacteria and indirectly lowering ARGs abundance. The negative correlations observed between yeasts, ARGs hosts, and ARGs themselves, along with the increased yeast abundance during fermentation, further support this inference (Fig. 6c).

Therefore, LAB inoculation not only improved fermentation quality but also indirectly suppressed the dissemination potential of ARGs by reshaping microbial interaction networks. Based on the PLS-PM results, it can be inferred that environmental and compositional factors first shape microbial community structure, which subsequently influences ARGs variation through changes in MGEs profiles. This hierarchical association pattern provides an ecological explanation for ARGs attenuation in silage and aerobic exposure systems. Previous studies have also highlighted the negative regulatory effects of environmental selection pressure on ARGs. For instance, Zhu et al. demonstrated that environmental stress could suppress or eliminate MGEs activity [57], thereby reducing ARGs dissemination potential. The present study further supports this mechanism, showing that environmental and compositional stresses may restrict ARGs propagation through microbial mediation and reduced MGEs-associated mobility potential. Overall, our findings indicate that ARGs dynamics are largely shaped by the microbial–MGEs coupling mechanism. Although horizontal gene transfer may represent an important potential route of ARGs dissemination, its contribution should be interpreted with caution because direct transfer events and host-resolved ARGs and MGEs linkages were not experimentally validated in this study. These results emphasize the pivotal ecological role of microbial communities in ARGs regulation under environmental and substrate constraints.

3.5. *Mechanisms of ARGs dissemination during aerobic exposure of cattail silage*

Based on the above results, we constructed a conceptual model to illustrate the mechanisms of ARGs dissemination during aerobic exposure of cattail silage. As environmental pressures increased and substrate availability declined, the abundance of MGEs reflecting ARGs mobility significantly decreased, thereby lowering the potential risk of ARGs transfer (Fig. 7a). These changes reflected a complex microbial response to aerobic stress, in which oxygen exposure induced both taxonomic and functional restructuring of the silage microbiome. Such shifts in community metabolism and interaction patterns ultimately shaped the

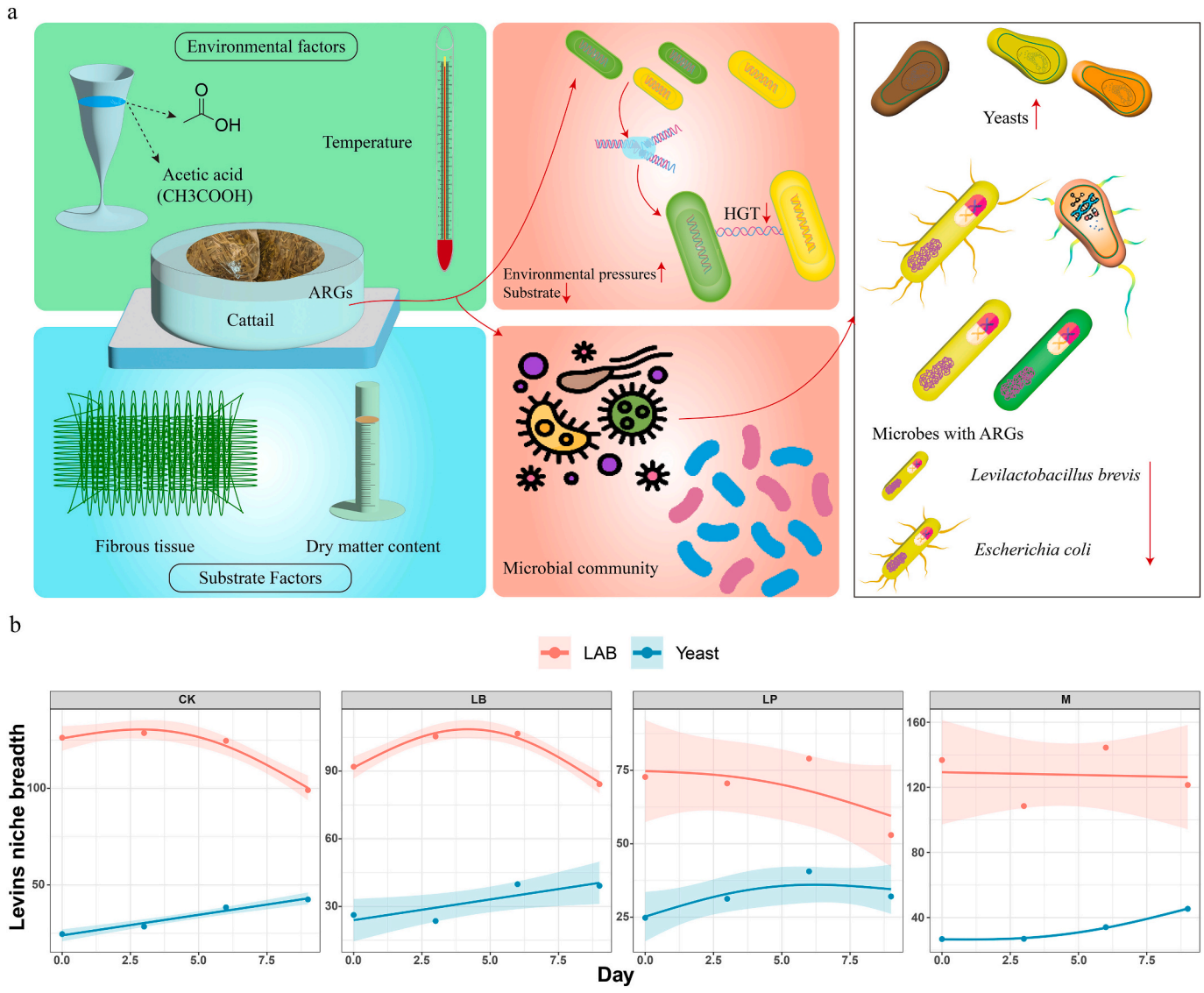


Fig. 7. Conceptual model illustrates microbial and environmental mechanisms driving the reduction of ARG abundance during aerobic exposure of cattail silage. (a) Mechanisms of reduced abundance of ARGs during aerobic exposure of cattail silage; (b) Changes in ecological niche breadth during aerobic exposure of cattail silage; LAB, lactic acid bacteria. CK, without inoculants; LB, *L. brevis*; LP, *L. plantarum*; M, a mixture of *L. brevis* and *L. plantarum*.

ecological processes influencing ARGs dissemination. As shown in Fig. 7b, the ecological niche breadth of lactic acid bacteria gradually narrowed over time, while that of yeasts expanded, suggesting that fungi progressively occupied ecological space previously dominated by bacteria. This expansion enabled yeast-like fungi to rapidly consume available substrates and diminish the growth advantage of ARGs-hosting bacteria, contributing to the observed decrease in ARGs abundance. Similar bacterial–fungal niche differentiation has been documented in the decomposition of plant-derived organic matter, where fungal proliferation suppresses bacterial abundance by occupying bacterial niches and competing for carbon resources [58,59]. In this study, such niche expansion and resource-driven microbial competition jointly reshaped community composition and reduced the ecological potential for ARGs dissemination during the later stages of aerobic exposure.

Acetic acid acts as an organic acid to lower pH and synergistically regulates horizontal ARGs transfer efficiency along with temperature. On the other hand, as a substrate utilized by certain bacteria [60], acetic acid regulates ARGs dissemination risks through competitive interactions. However, while serving as a bacterial substrate, acetic acid's antimicrobial properties also inhibit the activity of some yeasts [61]. This inhibitory effect led to growth fatigue in yeast-like fungi in the other three treatment (CK, LB and M) groups (Fig. 3, Table 1), resulting in high ARGs abundance. In the LP, although low levels of acetic acid were detected, its significant inhibitory effect on ARGs remained evident. LP significantly inhibited ARGs-carrying bacteria by promoting the proliferation of lactic acid bacteria. Particularly under aerobic exposure, LP fermentation produced to high levels of lactic acid, serving as a primary energy source for yeast-like fungi, promoting their proliferation and ecological niche occupation, further restricting the growth and proliferation of ARGs-carrying bacteria.

Overall, the key novelty of this study is that ARGs attenuation during aerobic exposure was linked not only to reduced MGEs-mediated mobility and suppression of bacterial hosts, but also to bacterial–fungal niche replacement driven by the expansion of yeast-like fungi. Specifically, our results reveal that microbial community dynamics influence ARGs dissemination during aerobic exposure of silage, and that yeast-like fungi may inhibit ARGs-carrying bacteria through competition for substrate resources and ecological niche occupation. This finding extends previous studies that mainly focused on bacterial hosts and MGEs by highlighting the potential role of fungal succession in regulating resistome dynamics during aerobic exposure. Future investigations should further clarify whether these yeasts directly carry ARGs or indirectly suppress ARGs by reshaping bacterial niches, substrate competition, and organic acid metabolism. These findings also suggest that future silage inoculant development should consider not only fermentation quality, but also ARGs-host suppression, MGEs activity, bacterial–fungal interactions, and resistome safety during aerobic handling.

4. Conclusion

Reduction of ARGs-related risks during aerobic exposure was linked to the coordinated effects of acidified microenvironments, host community restructuring, and reduced MGEs-associated mobility potential. Notably, the self-isolated LP inoculant outperformed other inoculation schemes in reducing the abundance of potential ARGs-hosting bacteria and lowering ARGs mobility-related risk. From a bioprocess perspective, targeted single-strain inoculation can be deployed to enhance bioresource safety during the most practical exposure stage in silage use. The present study provides an association-based framework for integrating microbial process control with antimicrobial resistance risk management in high-altitude forage systems.

CRediT authorship contribution statement

Chaosheng Liao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Mingjie Zhang:** Resources, Methodology, Investigation, Data curation. **Wenlong Dai:** Methodology, Investigation, Data curation. **Xiaolong Tang:** Methodology, Investigation. **Pan Wang:** Visualization, Software, Methodology. **Ning Mao:** Methodology, Investigation. **Ping Li:** Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Some icons used in Figs. 6 and 7 were downloaded from https://www.flaticon.com/.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.178822.

Data availability

Data will be made available on request.

References

[1] I.N. Okeke, M.E.A. de Kraker, T.P. Van Boeckel, C.K. Kumar, H. Schmitt, A.C. Gales, S. Bertagnolio, M. Sharland, R. Laxminarayan, The scope of the antimicrobial resistance challenge, Lancet 403 (2024) 2426–2438, https://doi.org/10.1016/S0140-6736(24)00876-6.

[2] C. Von Wintersdorff, J. Penders, J. Van Niekerk, N. Mills, S. Majumder, L. Van Alphen, P. Savelkoul, P. Wolffs, Dissemination of antimicrobial resistance in microbial ecosystems through horizontal gene transfer, Front. Microbiol. 7 (2016), https://doi.org/10.3389/fmicb.2016.00173.

[3] Z. Wu, X. Shao, Q. Wang, Antibiotics and antibiotic resistance genes in the environment: dissemination, ecological risks, and remediation approaches, Microorganisms (2025), https://doi.org/10.3390/microorganisms13081763.

[4] B. Han, Q. Li, J. Yu, W. Yang, M.G. Su, X. Hilal, S. Zhang Li, H. Li, The source, fate and prospect of antibiotic resistance genes in soil: a review, Front. Microbiol. 13 (2022), https://doi.org/10.3389/fmicb.2022.976657.

[5] D. Zheng, G. Yin, M. Liu, L. Hou, Y. Yang, T.P. Van Boeckel, Y. Zheng, Y. Li, Global biogeography and projection of soil antibiotic resistance genes, Sci. Adv. 8 (2022) eabq8015. https://www.science.org/doi/abs/10.1126/sciadv.abq8015.

[6] F. Cerqueira, V. Matamoros, J. Bayona, T. Berendonk, G. Elsinga, L. Hornstra, B. Piña, Antibiotic resistance gene distribution in agricultural fields and crops, A soil-to-food analysis. Environ. Res. 177 (2019) 108608, https://doi.org/10.1016/j.envres.2019.108608.

[7] F. Wang, W. Han, S. Chen, W. Dong, M. Qiao, C. Hu, B. Liu, Fifteen-year application of manure and chemical fertilizers differently impacts soil ARGs and microbial community structure, Front. Microbiol. 11 (2020), https://doi.org/10.3389/fmicb.2020.00062.

[8] S.M. Zainab, M. Junaid, N. Xu, R.N. Malik, Antibiotics and antibiotic resistant genes (ARGs) in groundwater: a global review on dissemination, sources, interactions, environmental and human health risks, Water Res. 187 (2020) 116455, https://doi.org/10.1016/j.watres.2020.116455.

[9] C. Liao, B. Na, X. Tang, M. Zhao, C. Zhang, S. Chen, M. You, B. Bai, L. Hao, D. Tondrob, G. Qu, S. Yang, B. Huang, W. Gou, Y. Xie, S. Bai, C. Chen, P. Li, Contribution of the bacterial community of poorly fermented oat silage to biogas emissions on the Qinghai Tibetan plateau, Sci. Total Environ. 897 (2023) 165336, https://doi.org/10.1016/j.scitotenv.2023.165336.

[10] O.C.M. Queiroz, I.M. Ogunade, Z. Weinberg, A.T. Adesogan, Silage review: foodborne pathogens in silage and their mitigation by silage additives, J. Dairy Sci. 101 (2018) 4132–4142, https://doi.org/10.3168/jds.2017-13901.

[11] D. Xu, X. Zhang, X. Yuan, H. Han, Y. Xue, X. Guo, Hazardous risk of antibiotic resistance genes: host occurrence, distribution, mobility and vertical transmission

from different environments to corn silage, Environ. Pollut. (2023) 122671, https://doi.org/10.1016/j.envpol.2023.122671.
[12] X. Zhang, Z. Ding, S. Usman, J. Zhang, M. Chen, X. Guo, Metagenomics insights into the effects of lactic acid bacteria inoculation on the biological reduction of antibiotic resistance genes in alfalfa silage, J. Hazard. Mater. 443 (2023) 130329, https://doi.org/10.1016/j.jhazmat.2022.130329.
[13] A. Tarrah, D. Zhang, P. Darvishzadeh, G. LaPointe, The contribution of dairy bedding and silage to the dissemination of genes coding for antimicrobial resistance: a narrative review, Antibiotics 13 (2024), https://doi.org/10.3390/antibiotics13090905.
[14] Y. Zhang, H. Hu, Q. Chen, B.K. Singh, H. Yan, D. Chen, J. He, Transfer of antibiotic resistance from manure-amended soils to vegetable microbiomes, Environ. Int. 130 (2019) 104912, https://doi.org/10.1016/j.envint.2019.104912.
[15] C.O. Okoye, Y. Wu, Y. Wang, L. Gao, X. Li, J. Jiang, Fermentation profile, aerobic stability, and microbial community dynamics of corn straw ensiled with Lactobacillus buchneri PC-C1 and Lactobacillus plantarum PC1-1, Microbiol. Res. 270 (2023) 127329, https://doi.org/10.1016/j.micres.2023.127329.
[16] L. Chen, S. Bai, M. You, B. Xiao, P. Li, Y. Cai, Effect of a low temperature tolerant lactic acid bacteria inoculant on the fermentation quality and bacterial community of oat round bale silage, Anim. Feed Sci. Technol. 269 (2020) 114669, https://doi.org/10.1016/j.anifeedsci.2020.114669.
[17] D. Xu, X. Zhang, S. Usman, J. Bai, N. Sheoran, X. Guo, Reducing transmission of high-risk antibiotic resistance genes in whole-crop corn silage through lactic acid bacteria inoculation and increasing ensiling temperature, Sci. Total Environ. (2024) 172114, https://doi.org/10.1016/j.scitotenv.2024.172114.
[18] X. Zhang, D. Xu, S. Usman, Y. Li, Y. Liang, J. Bai, X. Guo, Heterofermentative Lentilactobacillus buchneri and low dry matter reduce high-risk antibiotic resistance genes in corn silage by regulating pathogens and mobile genetic element, J. Hazard. Mater. 479 (2024) 135700, https://doi.org/10.1016/j.jhazmat.2024.135700.
[19] V. Rozman, M. Lorbeg, P. Treven, T. Accetto, S. Janežič, M. Rupnik, B. Matijašić, Genomic insights into antibiotic resistance and mobilome of lactic acid bacteria and bifidobacteria, Life Sci. Alliance 6 (2023), https://doi.org/10.26508/lsa.202201637.
[20] Z. Pei, F.A. Sadiq, X. Han, J. Zhao, H. Zhang, R.P. Ross, W. Lu, W. Chen, Comprehensive scanning of prophages in lactobacillus: distribution, diversity, antibiotic resistance genes, and linkages with CRISPR-Cas systems, mSystems 6 (2021), https://doi.org/10.1128/msystems.01211-20, 10–1128.
[21] T. De Evan, A.R. Musa, C. Marcos, J.S. Alao, E. Iglesias, F. Escribano, M. Carro, Ensiling Typha (Typha latifolia) forage with different additives for ruminant feeding, In Vitro Studies. Applied Sciences (2023), https://doi.org/10.3390/app13116546.
[22] M. Güllap, M. Tan, S. Severoğlu, H. Aktaş, A. Yazici, Determination of nutritional values of cattail (Typha latifolia L.) and common reed (Phragmites australis Cav.) silages, Turkish Journal of Range and Forage Science (2024), https://doi.org/10.51801/turkjrfs.1581274.
[23] E.W. Rice, P. Wang, A.L. Smith, L.B. Stadler, Determining hosts of antibiotic resistance genes: a review of methodological advances, Environ. Sci. Technol. Lett. 7 (2020) 282–291, https://doi.org/10.1021/acs.estlett.0c00202.
[24] J. Guo, J. Li, H. Chen, P.L. Bond, Z. Yuan, Metagenomic analysis reveals wastewater treatment plants as hotspots of antibiotic resistance genes and mobile genetic elements, Water Res. 123 (2017) 468–478, https://doi.org/10.1016/j.watres.2017.07.002.
[25] W. Shuo, H. Peishan, Z. Chao, Z. Wei, C. Xiaoyang, Z. Qing, Novel strategy to understand the aerobic deterioration of corn silage and the influence of Neolamarckia cadamba essential oil by multi-omics analysis, Chem. Eng. J. 482 (2024) 148715, https://doi.org/10.1016/j.cej.2024.148715.
[26] Y. Xie, H. Sun, C. Zhang, Q. Cheng, Y. Zheng, C. Wang, B. Xiao, P. Li, C. Chen, Ambient ultraviolet radiation: a new factor affecting anaerobic fermentation of oat and subsequent methane emissions, Bioresour. Technol. 355 (2022) 127243, https://doi.org/10.1016/j.biortech.2022.127243.
[27] G.A. Broderick, J.H. Kang, Automated simultaneous determination of Ammonia and Total amino acids in ruminal fluid and in vitro Media1, J. Dairy Sci. 63 (1980) 64–75, https://doi.org/10.3168/jds.S0022-0302(80)82888-8.
[28] F. Shea, C.E. Watts, Dumas method for organic nitrogen, Ind. Eng. Chem. Anal. Ed. 11 (1939) 333–334, https://doi.org/10.1021/ac50134a013.
[29] P.J. Van Soest, J.B. Robertson, B.A. Lewis, Methods for dietary Fiber, neutral detergent Fiber, and nonstarch polysaccharides in relation to animal nutrition, J. Dairy Sci. 74 (1991) 3583–3597, https://doi.org/10.3168/jds.S0022-0302(91)78551-2.
[30] T. Arthur Thomas, An automated procedure for the determination of soluble carbohydrates in herbage, J. Sci. Food Agric. 28 (1977) 639–642, https://doi.org/10.1002/jsfa.2740280711.
[31] B. Buchfink, C. Xie, D.H. Huson, Fast and sensitive protein alignment using DIAMOND, Nat. Methods 12 (2015) 59–60, https://doi.org/10.1038/nmeth.3176.
[32] D.H. Huson, S. Mitra, H. Ruscheweyh, N. Weber, S.C. Schuster, Integrative analysis of environmental sequences using MEGAN4, Genome Res. 21 (9) (2011) 1552–1560, https://doi.org/10.1101/gr.120618.111.
[33] A.G. McArthur, N. Waglechner, F. Nizam, A. Yan, M.A. Azad, A.J. Baylay, K. Bhullar, M.J. Canova, G. De Pascale, L. Ejim, L. Kalan, A.M. King, K. Koteva, M. Morar, M.R. Mulvey, J.S. O'Brien, A.C. Pawlowski, L.J.V. Piddock, P. Spanogiannopoulos, A.D. Sutherland, I. Tang, P.L. Taylor, M. Thaker, W. Wang, M. Yan, T. Yu, G.D. Wright, The comprehensive antibiotic resistance database, Antimicrob. Agents Chemother. 57 (2013) 3348–3357, https://doi.org/10.1128/aac.00419-13.
[34] B. Jia, A.R. Raphenya, B. Alcock, N. Waglechner, P. Guo, K.K. Tsang, B.A. Lago, B. M. Dave, S. Pereira, A.N. Sharma, S. Doshi, M. Courtot, R. Lo, L.E. Williams, J. G. Frye, T. Elsayegh, D. Sardar, E.L. Westman, A.C. Pawlowski, T.A. Johnson, F.S. L. Brinkman, G.D. Wright, A.G. McArthur, CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database, Nucleic Acids Res. 45 (2017) D566–D573, https://doi.org/10.1093/nar/gkw1004.
[35] J.L. Martínez, T.M. Coque, F. Baquero, What is a resistance gene? Ranking risk in resistomes, Nat. Rev. Microbiol. 13 (2015) 116–123, https://doi.org/10.1038/nrmicro3399.
[36] R. Zhao, K. Yu, J. Zhang, G. Zhang, J. Huang, L. Ma, C. Deng, X. Li, B. Li, Deciphering the mobility and bacterial hosts of antibiotic resistance genes under antibiotic selection pressure by metagenomic assembly and binning approaches, Water Res. 186 (2020) 116318, https://doi.org/10.1016/j.watres.2020.116318.
[37] X. Tang, C. Liao, S. Zhou, C. Chen, L. Li, G. Lu, X. Huang, M. Zhang, C. Chen, P. Li, Potential of perennial sorghum for biogas production: pretreatment with yeast-contained inoculants during anaerobic storage, Fuel 359 (2024) 130365, https://doi.org/10.1016/j.fuel.2023.130365.
[38] P. Li, W. Zhao, L. Yan, L. Chen, Y. Chen, W. Gou, M. You, Q. Cheng, C. Chen, Inclusion of abandoned rhubarb stalk enhanced anaerobic fermentation of alfalfa on the Qinghai Tibetan plateau, Bioresour. Technol. 347 (2022) 126347, https://doi.org/10.1016/j.biortech.2021.126347.
[39] W. Li, J. Zeng, N. Zheng, C. Ge, Y. Li, H. Yao, Polyvinyl chloride microplastics in the aquatic environment enrich potential pathogenic bacteria and spread antibiotic resistance genes in the fish gut, J. Hazard. Mater. 475 (2024) 134817, https://doi.org/10.1016/j.jhazmat.2024.134817.
[40] Z. Lu, J.A. Imlay, When anaerobes encounter oxygen: mechanisms of oxygen toxicity, tolerance and defence, Nat. Rev. Microbiol. 19 (2021) 774–785, https://doi.org/10.1038/s41579-021-00583-y.
[41] G. Wang, X. Gao, Y. Cai, G. Li, R. Ma, J. Yuan, Dynamics of antibiotic resistance genes during manure composting: reduction in herbivores manure and accumulation in carnivores, Environ. Int. 190 (2024) 108900, https://doi.org/10.1016/j.envint.2024.108900.
[42] Z. Yu, Z. Liu, L. Sun, C. Dong, Y. Jin, B. Hu, D. Cheng, Mobile genetic elements mediate the cross-media transmission of antibiotic resistance genes from pig farms and their risks, Sci. Total Environ. 926 (2024) 172115, https://doi.org/10.1016/j.scitotenv.2024.172115.
[43] K. Peng, L. Jin, Y.D. Niu, Q. Huang, T.A. McAllister, H.E. Yang, H. Denise, Z. Xu, S. Acharya, S. Wang, Y. Wang, Condensed tannins affect bacterial and fungal microbiomes and mycotoxin production during ensiling and upon aerobic exposure, Appl. Environ. Microbiol. 84 (2018) e02217–e02274, https://doi.org/10.1128/AEM.02274-17.
[44] Q. Wang, J. Gu, X. Wang, J. Ma, T. Hu, H. Peng, J. Bao, R. Zhang, Effects of nano-zerovalent iron on antibiotic resistance genes and mobile genetic elements during swine manure composting, Environ. Pollut. 258 (2020) 113654, https://doi.org/10.1016/j.envpol.2019.113654.
[45] P. Wang, Z. Qiao, X. Li, D. Wu, B. Xie, Fate of integrons, antibiotic resistance genes and associated microbial community in food waste and its large-scale biotreatment systems, Environ. Int. 144 (2020) 106013, https://doi.org/10.1016/j.envint.2020.106013.
[46] W.J. Middelhoven, A.H.M. van Baalen, Development of the yeast flora of whole-crop maize during ensiling and during subsequent aerobiosis, J. Sci. Food Agric. 42 (1988) 199–207, https://doi.org/10.1002/jsfa.2740420303.
[47] Z. Chen, Q. Fu, Q. Wen, Y. Wu, H. Bao, J. Guo, Microbial community competition rather than high-temperature predominates ARGs elimination in swine manure composting, J. Hazard. Mater. 423 (2022) 127149, https://doi.org/10.1016/j.jhazmat.2021.127149.
[48] M.C. Roberts, S. Schwarz, Tetracycline and Phenicol resistance genes and mechanisms: importance for agriculture, the environment, and humans, J. Environ. Qual. 45 (2016) 576–592, https://doi.org/10.2134/jeq2015.04.0207.
[49] F. Li, W. Ke, Z. Ding, J. Bai, Y. Zhang, D. Xu, Z. Li, X. Guo, Pretreatment of Pennisetum sinese silages with ferulic acid esterase-producing lactic acid bacteria and cellulase at two dry matter contents: fermentation characteristics, carbohydrates composition and enzymatic saccharification, Bioresour. Technol. 295 (2020) 122261, https://doi.org/10.1016/j.biortech.2019.122261.
[50] K. Li, R. Cao, S. Mo, R. Yao, Z. Ren, J. Wu, Swine manure composting with compound microbial inoculants: removal of antibiotic resistance genes and their associations with microbial community, Front. Microbiol. 11 (2020), https://doi.org/10.3389/fmicb.2020.592592.
[51] R. Cao, W. Ben, Z. Qiang, J. Zhang, Removal of antibiotic resistance genes in pig manure composting influenced by inoculation of compound microbial agents, Bioresour. Technol. 317 (2020) 123966, https://doi.org/10.1016/j.biortech.2020.123966.
[52] B. Li, Y. Yang, L. Ma, F. Ju, F. Guo, J.M. Tiedje, T. Zhang, Metagenomic and network analysis reveal wide distribution and co-occurrence of environmental antibiotic resistance genes, ISME J. 9 (2015) 2490–2502, https://doi.org/10.1038/ismej.2015.59.
[53] G. Zhou, X. Qiu, X. Wu, S. Lu, Horizontal gene transfer is a key determinant of antibiotic resistance genes profiles during chicken manure composting with the addition of biochar and zeolite, J. Hazard. Mater. 408 (2021) 124883, https://doi.org/10.1016/j.jhazmat.2020.124883.
[54] T. Heunis, S. Deane, S. Smit, L.M.T. Dicks, Proteomic profiling of the acid stress response in Lactobacillus plantarum 423, J. Proteome Res. 13 (2014) 4028–4039, https://doi.org/10.1021/pr500353x.
[55] C. Zhang, Z. Zhao, S. Dong, D. Zhou, Simultaneous elimination of amoxicillin and antibiotic resistance genes in activated sludge process: contributions of easy-to-

biodegrade food, Sci. Total Environ. 764 (2021) 142907, https://doi.org/10.1016/j.scitotenv.2020.142907.
[56] Y. Chen, P. Li, Y. Huang, K. Yu, H. Chen, K. Cui, Q. Huang, J. Zhang, K. Yew-Hoong Gin, Y. He, Environmental media exert a bottleneck in driving the dynamics of antibiotic resistance genes in modern aquatic environment, Water Res. 162 (2019) 127–138, https://doi.org/10.1016/j.watres.2019.06.047.
[57] L. Zhu, Y. Zhao, K. Yang, J. Chen, H. Zhou, X. Chen, Q. Liu, Z. Wei, Host bacterial community of MGEs determines the risk of horizontal gene transfer during composting of different animal manures, Environ. Pollut. 250 (2019) 166–174, https://doi.org/10.1016/j.envpol.2019.04.037.
[58] C. Wang, Y. Kuzyakov, Mechanisms and implications of bacterial–fungal competition for soil resources, ISME J. 18 (2024) wrae073, https://doi.org/10.1093/ismejo/wrae073.
[59] W.D. Boer, L.B. Folman, R.C. Summerbell, L. Boddy, Living in a fungal world: impact of fungi on soil bacterial niche development★, FEMS Microbiol. Rev. 29 (2005) 795–811, https://doi.org/10.1016/j.femsre.2004.11.005.
[60] G. Gong, B. Wu, L. Liu, J. Li, Q. Zhu, M. He, G. Hu, Metabolic engineering using acetate as a promising building block for the production of bio-based chemicals, Engineering Microbiology 2 (2022) 100036, https://doi.org/10.1016/j.engmic.2022.100036.
[61] R.E. Muck, E.M.G. Nadeau, T.A. McAllister, F.E. Contreras-Govea, M.C. Santos, L. Kung, Silage review: recent advances and future uses of silage additives, J. Dairy Sci. 101 (2018) 3980–4000, https://doi.org/10.3168/jds.2017-13839.